



Customer Information System using HBase & HiveQL

◆ a. Create Tables in Hive

Start Hive:

```
Bash
hive
```

Create Customer Table

```
SQL
CREATE TABLE customer_info(
cust_id INT,
cust_name STRING,
order_id INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ',';
```

Create Order Table

```
SQL
CREATE TABLE order_info(
order_id INT,
item_id INT,
quantity INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ',';
```

Create Item Table

```
SQL
CREATE TABLE item_info(
item_id INT,
item_name STRING,
item_price INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ',';
```

◆ b. Load Data from Local Storage

Create Sample Files

customer.txt

```
1,Rahul,101
2,Amit,102
3,Priya,103
```

order.txt

```
101,201,2
102,202,1
103,203,4
```

item.txt

```
201,Laptop,50000
202,Mouse,500
203,Keyboard,1000
```

Load Data

```
SQL
LOAD DATA LOCAL INPATH '/home/cloudera/customer.txt'
INTO TABLE customer_info;
```

```
SQL
LOAD DATA LOCAL INPATH '/home/cloudera/order.txt'
INTO TABLE order_info;
```

```
SQL
LOAD DATA LOCAL INPATH '/home/cloudera/item.txt'
INTO TABLE item_info;
```

◆ c. Perform Join in Hive

```
SQL
SELECT c.cust_name,
       o.order_id,
       i.item_name,
       o.quantity,
       i.item_price
FROM customer_info c
JOIN order_info o
ON c.order_id = o.order_id
JOIN item_info i
ON o.item_id = i.item_id;
```

◆ d. Create Index on Customer Table

```
SQL
CREATE INDEX idx_customer
ON TABLE customer_info(cust_name)
AS 'COMPACT'
WITH DEFERRED REBUILD;
```

Rebuild Index

```
SQL
ALTER INDEX idx_customer ON customer_info REBUILD;
```

◆ e. Find Total and Average Sales

Total Sales

```
SQL
SELECT SUM(o.quantity * i.item_price) AS total_sales
FROM order_info o
JOIN item_info i
ON o.item_id = i.item_id;
```

Average Sales

```
SQL
SELECT AVG(o.quantity * i.item_price) AS average_sales
FROM order_info o
JOIN item_info i
ON o.item_id = i.item_id;
```

◆ f. Find Order Details with Maximum Cost

```
SQL
SELECT c.cust_name,
       o.order_id,
       i.item_name,
       (o.quantity * i.item_price) AS total_cost
FROM customer_info c
JOIN order_info o
ON c.order_id = o.order_id
JOIN item_info i
ON o.item_id = i.item_id
ORDER BY total_cost DESC
LIMIT 1;
```

◆ g. Create External Hive Table Connected to HBase

Start HBase:

```
Bash
hbase shell
```

Create HBase Table

```
Bash
create 'customer_hbase', 'info'
```

Insert Data

```
Bash
put 'customer_hbase', '1', 'info:name', 'Rahul'
put 'customer_hbase', '1', 'info:order', '101'

put 'customer_hbase', '2', 'info:name', 'Amit'
put 'customer_hbase', '2', 'info:order', '102'
```

Open Hive again:

```
Bash
hive
```

External Hive Table

```
SQL
CREATE EXTERNAL TABLE customer_hive(
  cust_id STRING,
  name STRING,
  order_id STRING
)

STORED BY 'org.apache.hadoop.hive.hbase.HBaseStorageHandler'

WITH SERDEPROPERTIES (
  "hbase.columns.mapping"=":key,info:name,info:order"
)

TBLPROPERTIES ("hbase.table.name"="customer_hbase");
```

◆ h. Display Records of Customer Table in HBase

```
Bash
scan 'customer_hbase'
```



Useful Commands

Show Hive Tables

```
SQL
show tables;
```

Show HBase Tables

```
Bash
list
```

Describe HBase Table

```
Bash
describe 'customer_hbase'
```

Exit Hive

```
SQL
quit;
```

Exit HBase

```
Bash
exit
```

